CSEC PHYSICS

Jan 2021 – Paper 2

Solutions

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SECTION A

Answer ALL questions.

1. Figure 1 shows the apparatus which was used by two students to carry out an experiment to determine the focal length, f, of a lens by using the object distance, u, and the image distance, v.

   At the end of the experiment, the object distance, u, and the image distance, v, were recorded in Table 1.

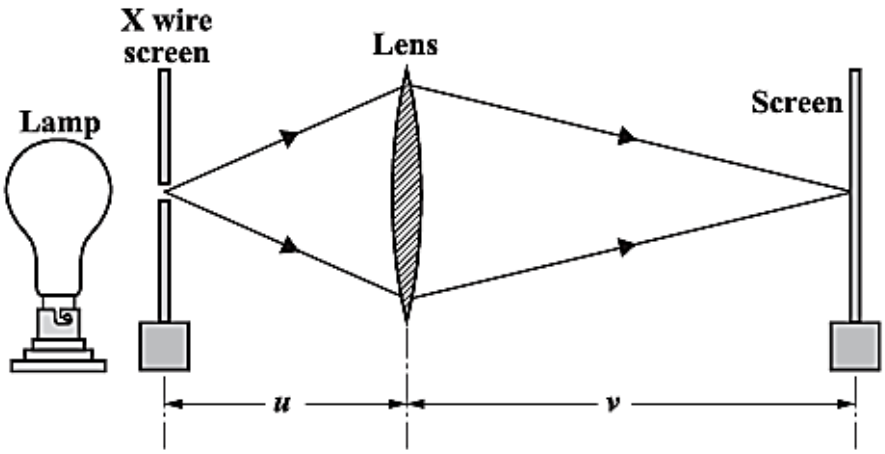

Figure 1: Experiment to measure the focal length, f, of a lens

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STUDENT HUB

TABLE 1: RESULTS OF THE EXPERIMENTS

u (cm)	$1/u(cm^{-1}) \times 10^{-2}$	v (cm)	$1/v(cm^{-1}) \times 10^{-2}$
20.0	5.0	60.0	1.7
30.0	3.3	30.0	3.3
35.0	2.9	26.3	3.8
40.0	2.5	24.0	4.2
45.0	2.2	22.5	4.4
55.0	1.8	20.6	4.9

(a) Complete Table 1 by calculating the missing values. (6 marks)

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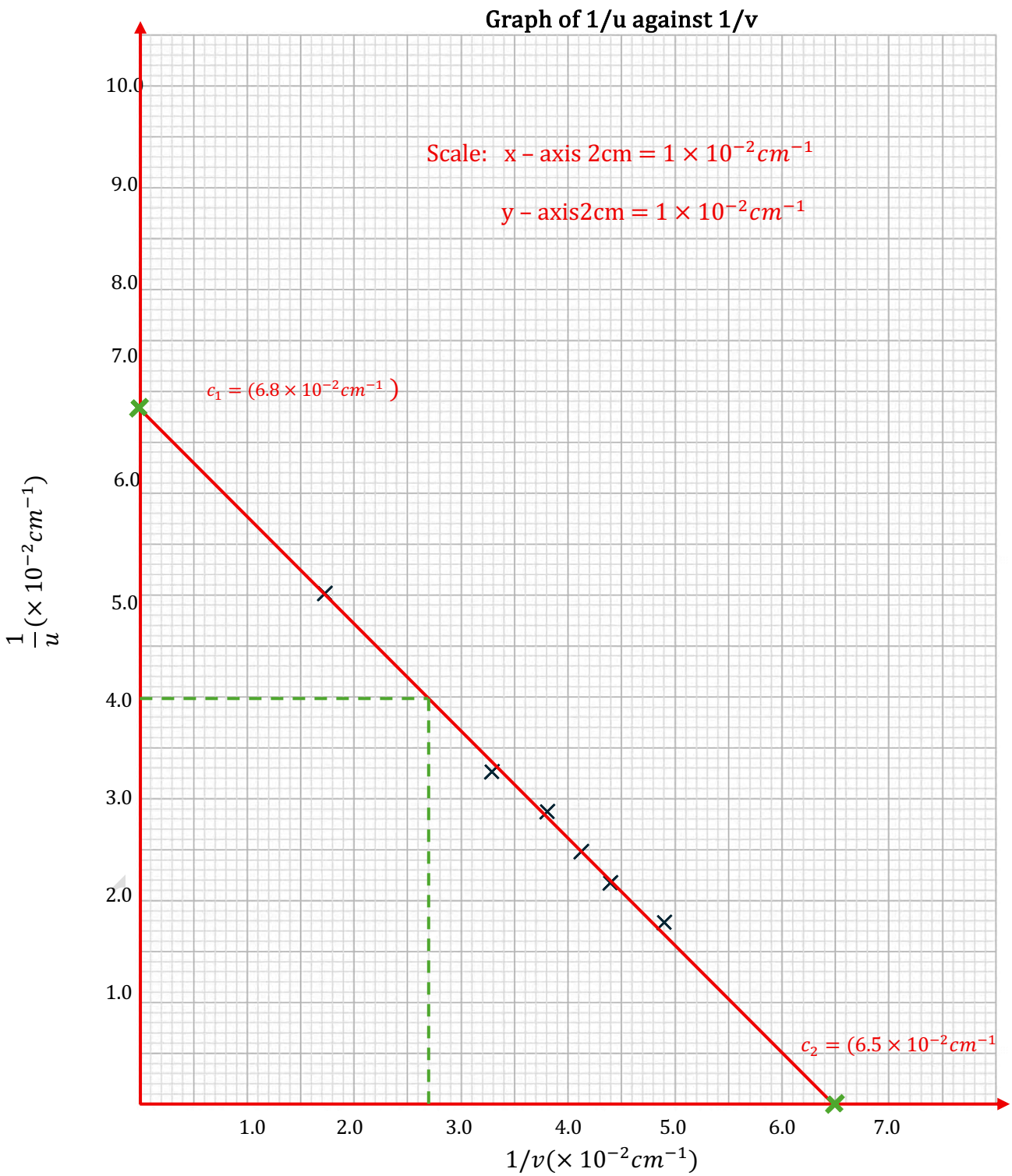
(b) On Figure 2, complete the graph of 1/u against 1/v using 2cm to represent $1 \times 10^{-2} cm^{-1}$ on each axis. (7 marks)



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(c) Use the graph to determine the value of v when 1/u is $4.0 \times 10^{-2} cm^{-1}$

$$\frac{1}{v} = 2.7 \times 10^{-2} cm^{-1}$$

$$v = \frac{1}{2.7 \times 10^{-2}}$$

$$v = 37.04\ cm\ to\ 2dp$$ (1 mark)

(d) (i)Use the graph to determine the intercepts, $c_1 and\ c_2$, on both axes , and calculate the value , c.

$$c_{avg} = \frac{(c_1 + c_2)}{2}$$

$$= 6.65 \times 10^{-2} cm^{-1}$$ (4 marks)

(ii) Use the formula c = 1/f to calculate the length, f, of the lens.

$$c = \frac{1}{f}$$

$$f = \frac{1}{c}$$

$$= \frac{1}{6.65 \times 10^{-2}}$$

$$f = 15\text{cm}$$ (3 marks)

(e) Complete the following statement:

The principal focus of a concave lens is defined as the _____point_____ on the principal axis from which rays originally _____parallel_____- to the principal axis _____diverge_____ after passing through the lens. (3 marks)

Total 25 marks

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2. (a) State the 'principle of moments' (3 marks)

The principle of moments states that for equilibrium, the sum of clockwise moments must be equal to the sum of anticlockwise moments about a point.

(b) State another condition that is necessary for a body to be in equilibrium.

Net forces = 0, for example, sum of upward forces = sum of downward forces

(c) State the formula for the moment of a force, providing the meaning of each variable and its SI unit. (3 marks)

$T = F \times d$

where T = moment of the force (Nm)

F = force (N)

d = distance from the pivot to the line of action of the force (m)

(d) Figure 3 shows a girl standing on a uniform plank, AB, of length 3 m and weight 270 N. The plank rests on two scales that are located at P and Q, a distance of 2 m apart. The scales are equidistant from the ends of the plank as shown in Figure 3. The scale at P reads 354 N and the scale at Q reads 417 N.

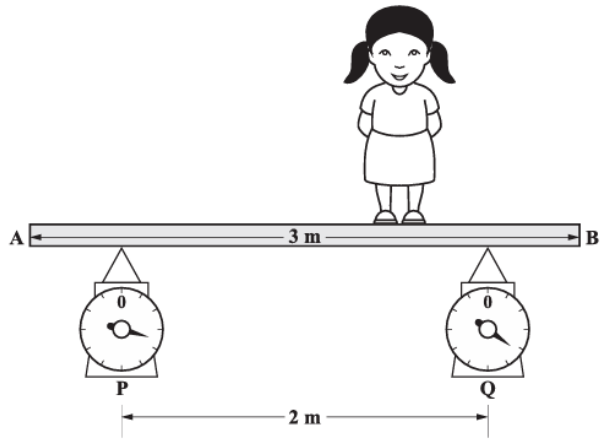

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Figure 3. A girl standing on a uniform plank

(i) On Figure 3, draw and label ALL the forces that are acting on the plank.

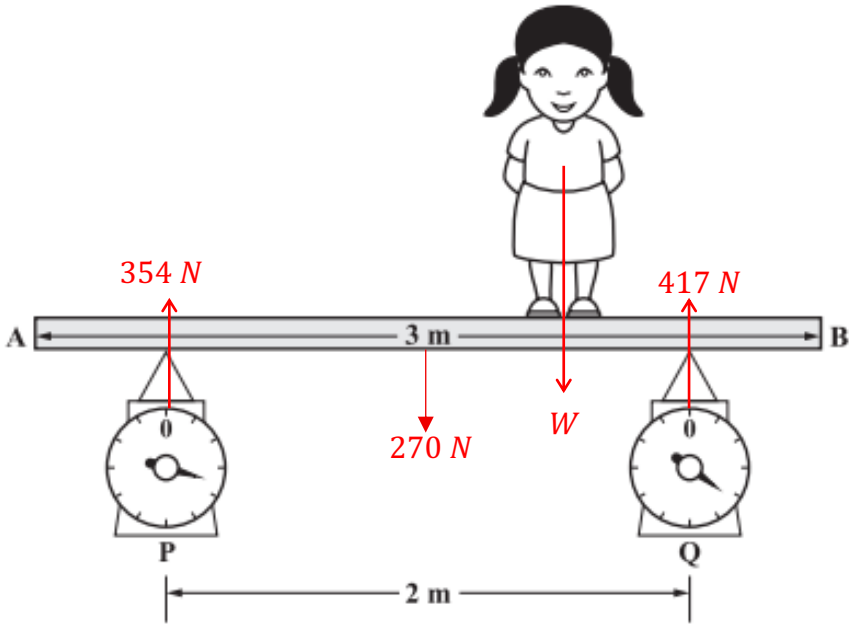

Figure 3. A girl standing on a uniform plank

(ii) Calculate the girl's weight.

Sum of upward forces = Sum of downward forces

$$354 + 417 = 270 + W$$

$$W = 354 + 417 - 270$$

$$W = 501\ N$$

∴ The girl's weight is $501\ N$.

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(e) By taking moments about Q, calculate the girl's distance from the scale at Q.

Taking moments about Q,

Total anticlockwise moments = Total clockwise moments

$$(501 \times d) + (270 \times 1) = (354 \times 2)$$

$$501d + 270 = 708$$

$$501d = 708 - 270$$

$$501d = 438$$

$$d = \frac{438}{501}$$

$$d = 0.87\ m$$

Total 15 marks

3. (a) (i) Define the term pressure. (2 marks)

Pressure is the normal force acting on a surface per unit area.

$$P = \frac{F}{A}$$

where P = pressure (Pa)

F = force (N)

A = area (m^2)

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(ii) State the SI unit of pressure pascal (Pa), in terms of its base units

The unit of Pressure $= \frac{Unit\ of\ force}{Unit\ of\ area}$

The unit of Pressure $= \frac{N}{m^2}$

The unit of Pressure $= \frac{kgms^{-2}}{m^2}$

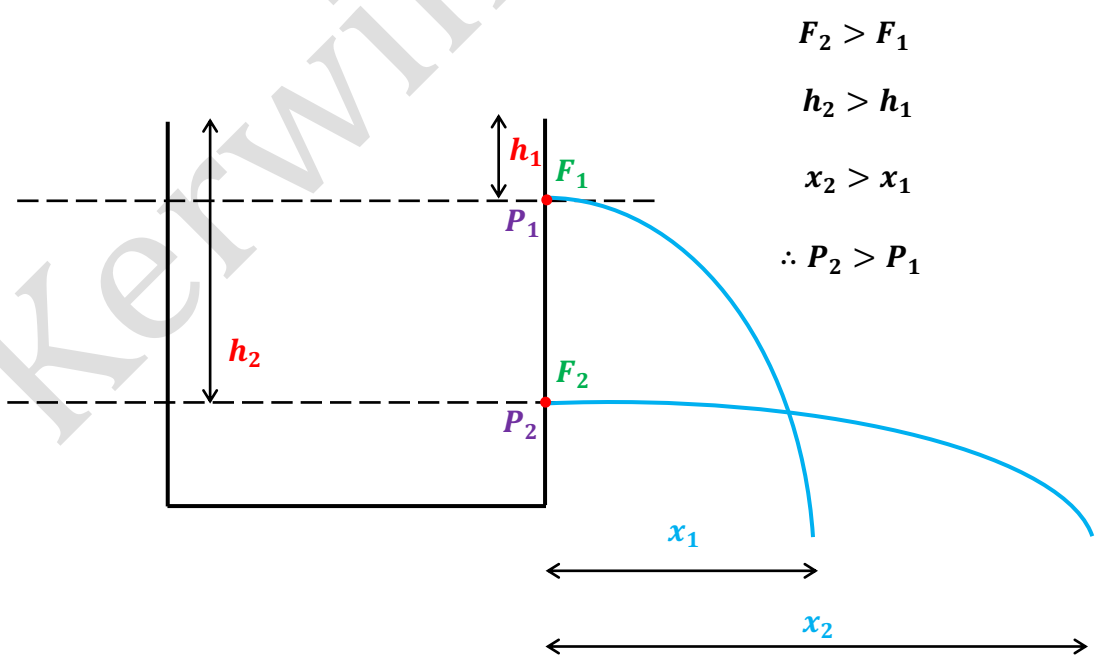
The unit of Pressure $= kgm^{-1}s^{-2}$

(iii) Centre of gravity (2 marks)

The point in a body through which the entire weight of the body appears to act

(b) (i) Describe, with the aid of a diagram, how the pressure in a liquid varies with the depth of the liquid. (3 marks)

The diagram is as follows:



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As the depth increases, the pressure also increases due to an increase in the weight of the fluid over that point. The weight of the fluid is dependent on the density of the fluid and the volume of the fluid. The volume above the fluid is proportional to the height of liquid over the point.

The force on the escaping liquid at P_2 is greater than P_1 and is visibly seen by the $x_2 > x_1$.

Hence, $P_2 > P_1$, since P_2 has a greater depth.

(ii) State the equation for the pressure difference (ΔP) between two points in a liquid, which are a vertical distance, h, apart.

The equation is as follows:

$$\Delta P = \rho g \Delta h$$

(c) Figure 4 shows a U – tube containing two immiscible liquids, X and Y.

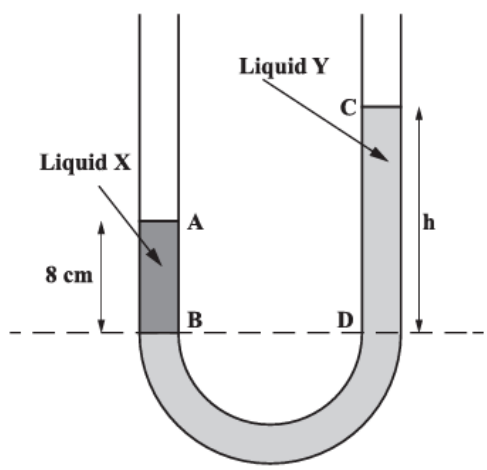

Figure 4. U – tube of immiscible liquids

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(i) If the density of Liquid X is 1350 $kg\ m^{-3}$, calculate the pressure (in Pa), due to the liquid column AB, which is 8 cm long. (3 marks)

$$P = \rho gh$$

$$P = 1350 \times 10 \times 0.08$$

$$P = 1080\ Pa$$

(ii) Hence, use your answer to (c) (i) to calculate the height, h, of the liquid column CD, given that the density of liquid Y is 900 $kg\ m^{-3}$. (State any equations used and fully explain the reasoning in obtaining your answer to (c) (ii).)

$$[\boldsymbol{Use\ g\ = 10ms^{-2}}]$$

Pressure in liquids are equal at the same depth.

Pressure at B = Pressure at D

$$1080 = \rho gh$$

$$1080 = 900 \times 10 \times h$$

$$h = \frac{1080}{9000}$$

$$= 0.12\ m$$

$$= 12\ cm$$

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SECTION B

Answer ALL questions.

4. (a) A young student has just entered secondary school. He was accustomed to having his home-cooked meal warmed up at lunch time in the microwave oven at his primary school. He misses this and suggests to his parents that they buy him a vacuum flask similar to the one shown in Figure 5.

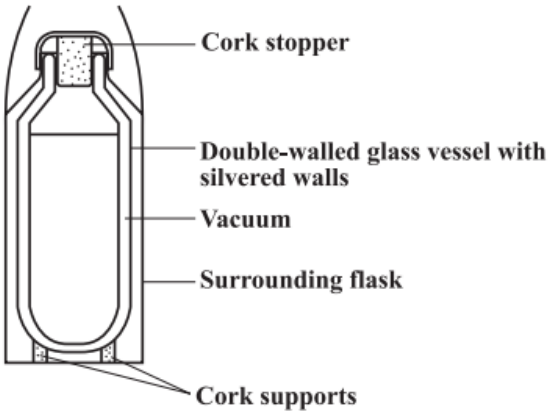

Figure 5. A vacuum flask

Using the information in Figure 5, explain in terms of the principle of thermal energy transfer how EACH of the following sections of the vacuum flask helps to keep the home-cooked meal warm. (6 marks)

(i) The cork stopper

- Cork is a poor conductor of heat, so it reduces heat loss by conduction.
- It also prevents convection currents by sealing the top, so warm air cannot escape.

(ii) The vacuum

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- A vacuum contains no particles, so heat cannot be transferred by conduction or convection.
- This greatly reduces heat loss from the liquid.

(iii) The double-walled glass vessel with silvered walls

- The glass walls are poor conductors, so they reduce heat loss by conduction.
- The silvered (shiny) surfaces reflect infrared radiation back towards the liquid, reducing heat loss by radiation.

(b) In an experiment to determine the specific latent heat of fusion of ice, shown in Figure 6, the following results were obtained.

The initial mass of the water is 120 g. The initial temperature of the water is 35 $°C$ and the initial temperature of the ice is 0 $°C$.

The final mass of the water is 140 g and the final temperature of the water plus the melted ice is 18 $°C$.

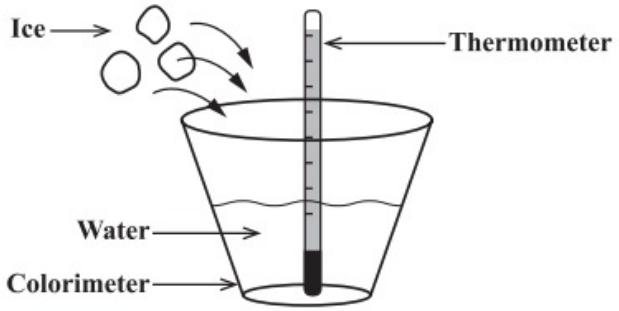

Figure 6. Experiment to determine the specific latent heat of fusion of ice

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Assume the container has negligible heat capacity and that there is no net heat transfer with the surroundings during the experiment.

(i) Determine the mass of the melted ice. (1 mark)

Mass = Final Mass – Initial mass

= 140 g – 120 g

= 20g

(ii) Calculate the heat lost by the water. (3 marks)

$Initial\ mass = 120$

$c = 4.2Jg^{-1}\,{}^{0}C^{-1}$

$\Delta T = 35 - 18 = 17^{0}C$

$Q = mc\Delta T$

$= (120)(4.2)(17)$

=8568J

(iii) Write an expression for the TOTAL heat gained by the ice in melting and the melted ice in warming to 18 $°C$. (2 marks)

Heat Energy lost from water = Heat energy gained by ice to water at 0^{0}C +Heat energy required to change from 0^{0}C to 18^{0}C

$Q_{total} = ml_f + mc\Delta T$

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(iv) Hence calculate the specific latent heat of fusion of ice. [3]

[Specific heat capacity of water = $4.2\ Jg^{-1}K^{-1}$]

$Q_{total} = ml_f + mc\Delta T$

$8568 = (20)l_f + (20)(4.2)(18)$

$8568 = 20l_f + 1512$

$l_f = \frac{8568-1512}{20}$

$= 352.8\ Jg^{-1}$

Total: 15 marks

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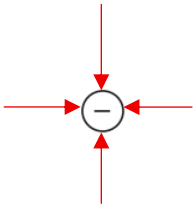
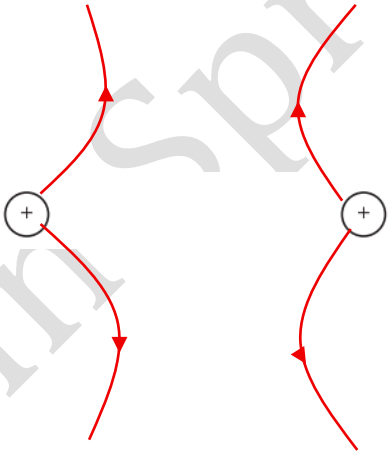
5. (a) Draw the electric field for EACH of the following arrangements.

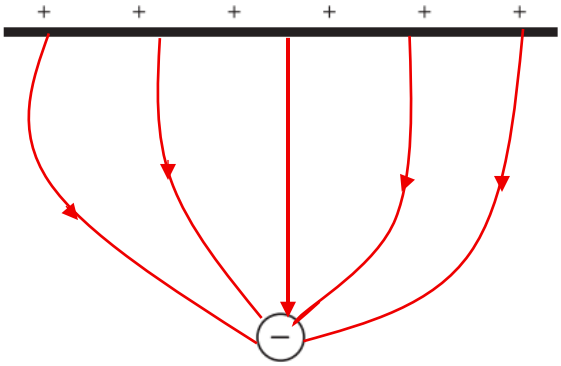
(i) [2]



(ii) [2]



(iii) [2]



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(b) If the number of electrons transferred from a charged rod to earth through a wire is 4×10^{18} and the total charge transferred is $0.64\ C$, calculate

(i) the magnitude of the charge on one electron [3]

$$0.64C = 4 \times 10^{18}x \ \ (where\ x\ is\ the\ charge\ of\ a\ single\ electron)$$

$$x = \frac{0.64}{4 \times 10^{18}}$$

$$= 1.6\ \times 10^{-19}C$$

(ii) the current that flows through the wire if it takes $0.40\ s$ for all the electrons to flow to earth (1 mark)

$t = 0.4s$

$Q = 0.64$

$Q = It$

$I = \frac{Q}{t}$

$= \frac{0.64}{0.40}$

$I = 1.6A$

(iii) State the direction in which the charges will flow, if the rod is positively charged. (1 mark)

Flows from the Earth to the rod

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(c) Explain how the static 'build up' of charges on a vehicle could be hazardous. [2]

The static build-up of charge on a vehicle can be hazardous because it can suddenly discharge as a spark when the vehicle touches a conductor or a person.

This spark can ignite flammable fuel vapours at places like petrol stations, causing a fire or explosion

Total: 15 marks

6. (a) Figure 7 shows the experimental arrangement of the apparatus used to carry out the Rutherford's alpha (α) particle scattering experiment to identify the structure of the atom.

A beam (collimated beam) of alpha particles from a radioactive source was fired at a thin piece of gold foil and the angle of deviation of the alpha particles observed.

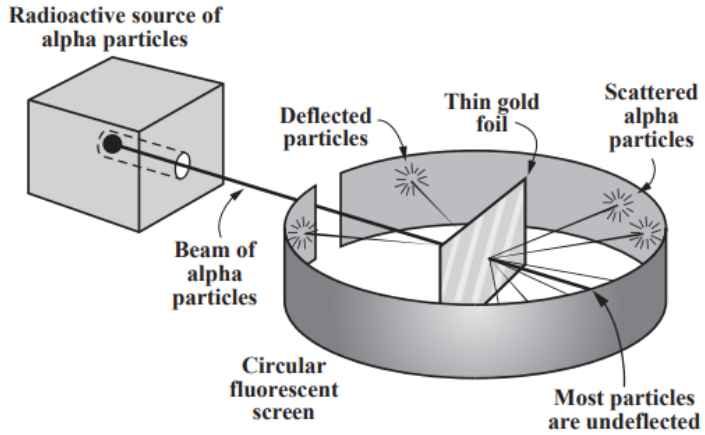

Figure 7. Rutherford's alpha particle scattering experiment

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(i) Write the names of the two scientists who assisted Rutherford in performing this experiment. [2]

Hans Geiger and Ernest Marsden

(ii) Explain why the experiment was carried out in an evacuated chamber.

The experiment was carried out in an evacuated chamber because air molecules would otherwise absorb or scatter the alpha particles, reducing their energy and changing their paths.

Removing air ensures the alpha particles travel unobstructed to the gold foil, so the results are accurate (2 marks)

(b) State the conclusions that were deduced from EACH of the following observations made from the alpha particle scattering experiment.

(i) Most of the alpha particles passed through the gold foil undeviated. (1 mark)

This shows that the atom is mostly empty space.

(ii) A few alpha particles were deflected from their path but continued through the gold foil. (1 mark)

This shows that there is a positively charged region in the atom that repels alpha particles (positive charge).

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(iii) A smaller number of alpha particles rebounded. (1 mark)

This shows that there is a small, dense, massive, positively charged nucleus at the centre of the atom.

(c) Describe the Rutherford model of the atom, based on the conclusions of the experiment stated in (b). (2 marks)

- The atom consists of a small, dense, positively charged nucleus at the centre.
- Most of the atom is empty space.
- Electrons move around the nucleus in the surrounding space.

(d) Energy is produced within stars by the process of fusion where alpha particles help convert beryllium into carbon, according to the equation below.

$$^{4}_{2}He + ^{9}_{4}Be \rightarrow ^{1}_{0}n + ^{12}_{6}C$$

(i) In the equation above, which symbol is used to represent the alpha particle? (1 mark)

$^{4}_{2}He$

(ii) The following table shows the mass of each nuclei in the process above.

$^{4}He_{2}$	$4.00260\ u$
$^{9}Be_{4}$	$9.01212\ u$
$^{1}n_{0}$	$1.00867\ u$
$^{12}C_{6}$	$12.0000\ u$

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Calculate the loss in mass, Δm, in kg, due to the reaction.

$$[\mathbf{1\ u = 1.66 \times 10^{-27}\ kg}]$$ (2 marks)

Mass defect $= Mass\ of\ Reactants - Mass\ of\ Products$

$= (4.00260 + 9.01212) - (12.0000 + 1.00867)$

$= 13.01472 - 13.00867$

$= 0.00605u\ or\ 6.05 \times 10^{-3}u$

$Mass\ defect\ =\ 0.00605 \times 1.66 \times 10^{-27}$

$= 1.0043 \times 10^{-29}$ kg

(iii) Calculate the energy equivalence of the mass in (d)(ii).

$$[\boldsymbol{c = 3.00 \times 10^{8}\ ms^{-1}}]$$

$E = mc^2$

$= (1.0043 \times 10^{-29})(\mathbf{3.00 \times 10^{8}})^2$

$= 9.04 \times 10^{-13} J\ to\ 3\ s.f$

(3 marks)

Total: 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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